Ajay Shetty

Senior Software Engineer — Backend • Cloud • Distributed systems

Scaler Profile — **8660187964** — **jyshtty@gmail.com** — **LinkedIn** — **GitHub** — **LeetCode** (3 star)

Summary

Lead software engineer with experience in designing distributed systems, building cloud-native platforms, and developing infrastructure on AWS. Specializes in platform engineering, event-driven architectures, secure data systems, and Agentic AI/GenAI-powered developer.

Skills

Languages: Python, Java, Go, JavaScript
Agentic AI: LangGraph, LangChain, RAG, MCP.
Databases: PostgreSQL, SQL, NoSQL
CI/CD: Jenkins, GitHub Actions, CodePipeline
Observability: Datadog, Grafana, CloudWatch
Frameworks: Django, FastAPI, Spring Boot, React.
Cloud: AWS, GCP, Docker, CloudFormation, IaC
Messaging: RabbitMQ, Kafka
Other: System Design, DSA, Data Engineering, Airflow, Kubernetes

Education

Woolf European University **Transcript link**
M.S. in Computer Software Engineering

BMS IT, Bangalore
B.E. in Electronics and Communication Engineering

Experience

EPAM Systems **Lead Engineer - Distributed Systems**
Client - Goldman Sachs. April 2026 – Present

- **Led architectural redesign of AML/KYC**, decoupling ACM into independent **Kubernetes pods** and integrating **Kafka**, **MongoDB**, and **Dell S3** to offload high-volume alert processing from AML/KYC application servers handling **1B+ transactions** and **100K+ alerts per batch run**.

Eli Lilly **Data Platform - Distributed Systems**
Global top 10 - $1T Valuation. Aug 2023 – April 2026

- Decomposed monolithic **notification and insurance claim processing** platform into an high-throughput **event-driven SOA architecture** (**API event trigger → APScheduler → RabbitMQ → SOA workers → RPC calls**), enabling **independent scaling** of **batch and real-time processing** across **5,000–10,000 daily transactions**.
- Architected and owned a production **LangGraph multi-agent AI system** serving **100+ engineers**, enabling **natural-language-driven IAM and database provisioning** across **72 research data domains with petabyte-scale**; integrated **FAISS-backed RAG** and a **human-in-the-loop approval gate** auto-escalating **30% of high-privilege requests** — governing **500+ access grants** and reducing **time-to-access from 3–5 days to under 10 minutes**.
- Contributed to **Research Data Platform** using **Spring Boot** and **Neo4j**-backed **data catalog** and **lineage tracking system** that manages **research data assets**, their versions, schemas, and provenance relationships through **gRPC APIs**.
- Delivered **SOLID-compliant** backend services using **multiple design patterns** to enable **encrypted Oracle connectivity** via **wallet-based authentication**.
- Architected a **multi-tenant PostgreSQL FDW platform** federating **47 database instances across 5 engines** into a **unified query interface** — eliminating **2-hour batch ETL lag** for real-time cross-database access, eliminating **recurring password failures** via automated credential rotation, removing pipeline maintenance overhead, and reducing **service downtime to zero**.
- **Built** a **fault-tolerant observability and alerting pipeline** using **CloudWatch**, **SNS**, **Lambda**, and **CloudFormation** to monitor 200+ database instance logs reducing **MTTR by 40%**.
- Integrated **schema validation** using **Pydantic**, reducing **errors caused by malformed inputs by over 30%**.

Amadeus IT Group **Software Development Engineer II – Distributed Systems**
World's largest GDS with over 4 billion transaction annually. Jan 2022 – Aug 2023

- **Led components of a high-throughput affiliate deeplinking and e-commerce tracking platform**, supporting **5k QPS** (bursts up to **15k+ QPS**), processing **120k+ deeplinks/month** and **50k+ transactions/month** across **50+ airline partners**, tracking reliability using first- and third-party cookies.
- Implemented **airline web scrapers** to extract routes and price details and feed normalized parameters into deeplink generation for affiliate sites - **Google Flights, Expedia, Skyscanner, Kayak**.

- **Designed an adapter-based integration architecture** supporting **25+ airline booking systems**, reducing per-partner code from **800+ LOC to 300 LOC**.
- **Implemented Protobuf-based message serialization**, reducing payload size by **65%** compared to JSON and enabling **3× faster response times** for partners (**Etihad, Qatar Airways etc**).

Zeomega Senior Software Developer – Distributed Systems
Patient Health Management - US Healthcare. Apr 2017 – Jan 2022

- **Built event driven data processing and rule-execution engine** processing **10M+ events/day**, improving **p95 latency by 35%** through **AST-based execution, shortcircuiting, and race-condition elimination**.
- **Engineered** production-grade **RabbitMQ AMQP infrastructure** processing **12k–18k messages/hour**, implementing publishers and consumers with **heartbeat and blocked-connection handling**, using the **Singleton design pattern** to prevent memory leaks, automating broker provisioning (virtual hosts, exchanges, queues) via **Python and shell scripts**, and upgrading **Pika to TLS 1.2/SSL** for secure messaging.
- **U**tilized **SQLAlchemy ORM** for efficient database interactions, ensuring data integrity for modules like patient records, document management (including SharePoint integration), and billing systems.
- **Built UI and API automation** using **Node.js**, **Protractor**, and **Jasmine**.

Projects

Event-Driven Architecture Framework

- **Designed** an **event-driven system** using the **Observer design pattern** and **decorators** to dynamically inject **SQL parameters** from API requests.
- **Eliminated injection risks** by replacing **eval()** with **AST-based evaluation**, preventing unsafe code execution.
- **Prevented infinite event loops** and **resolved race conditions** using custom decorators and **semaphores**.
- **Improved query execution performance** by **30%** using optimized logic and **Python short-circuiting**.

Service-Oriented Architecture Framework

- **Built** a **configurable SOA framework** enabling users to define custom execution rules for APIs.
- Supported both **synchronous** and **asynchronous XML-RPC calls**, enabling dynamic runtime control.
- **Reduced remote API invocation latency** by **25%** through efficient routing and caching.

Job Scheduling Microservice

- **Developed** a **modular job scheduler microservice** using **Python APScheduler**.
- Leveraged both **ThreadPoolExecutor** and **ProcessPoolExecutor** to **improve parallel task throughput** by **40%**.
- **Reduced queuing delays** under high load and implemented **observability**, execution logging, and failure recovery mechanisms.